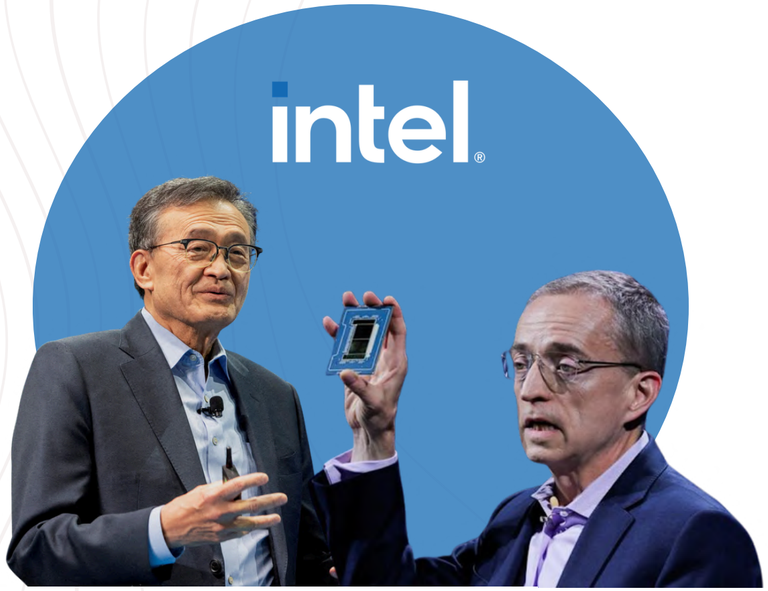


Group:

Marketing Analysis

Case Background



Core Business & Missed Opportunites

World's Leading semiconductor company, which built dominance through the "Intel Inside" campaign, supplying chips to companies such as Dell, HP, etc.

Missed Smart Revolution

- Qualcomm and Samsung dominated the mobile chips.

SAMSUNG QUALCOMM

- Intel had no pressure in the fastest-growing consumer tech market of the 2000s.

Falling Behind in GPUs



- Nvidia tripled in stock value between 2016 and 2017, driven by AI and deep learning demand.
- Intel's stock only grew 8% in the same period, clearly losing the next wave of computing.

Needed New Growth Frontier

- Smartwatches sit within the broader Internet of Things category.

Sensors + Connectivity

Exactly what Intel's chip expertise is built for

- Basic Science in 2014 acquisition in 2014 was Intel testing this new direction.

Timeline

2014

Basic Science

- **Acquired Basic Science** (Smartwatch company) with advanced heart rate sensors and sleep tracking platforms.

2015

Basis Peak

- **Launched the Basis Peak** smartwatch, emphasizing sensor quality and waterproofing

2016

Overheating

- **Battery Overheating** Caused Burns
- Full Product recall and Refunds
- Major reputational damage

Marketing Analysis

Data Processing and Analysis

1 Data Preparation

Original Dataset

Filter

ConstCom	TimelyInf	TaskMgm	DeviceSt	Wellness	Athlete	Style	AmznP	Female	Degree	Income	Age
3	2	3	3	2	3	3	1	1	1	2	38
6	6	6	6	5	3	1	1	0	2	3	38
7	4	4	4	6	4	1	0	0	1	3	42
7	5	4	5	5	4	4	1	0	2	5	35
7	4	2	6	3	2	4	1	0	1	3	36
2	1	3	2	2	4	3	0	1	2	2	47
7	7	3	6	3	1	4	1	1	1	5	36
7	5	6	5	2	2	3	1	0	1	3	41
6	5	6	3	5	3	5	1	0	1	2	46
5	6	6	3	5	5	3	1	0	1	3	28

Original Variables

- ***Constant Communication***
- ***Timely Information***
- ***Task Management***
- ***Device Sturdiness***
- ***Wellness***
- ***Athlete***
- ***Style***
- ***Amazon Prime***
- ***Female***
- ***Degree***
- ***Income***
- ***Age***

Segmentation Variables

	ConstCom	TimelyInf	TaskMgm	DeviceSt	Wellness	Athlete	Style
1	3	2	3	3	2	3	3
2	6	6	6	6	5	3	1
3	7	4	4	4	6	4	1
4	7	5	4	5	5	4	4
5	7	4	2	6	3	2	4
6	2	1	3	2	2	4	3
7	7	7	3	6	3	1	4
8	7	5	6	5	2	2	3
9	6	5	6	3	5	3	5
10	5	6	6	3	5	5	3

DESCRIPTION

Reason of Selected Variables

They capture the customers' needs and preferences - the basis for benefit segmentation.

1,000 ***Rows (n)***

7 ***Variables***

Standardized

	ConstCom	TimelyInf	TaskMgm	DeviceSt	Wellness	Athlete	Style
1	-1.0769876	-1.4623255	-0.7842874	-0.5352531	-1.4504902	-0.5490981	-1.0588291
2	0.8348247	1.0919548	1.1929077	1.3763651	0.3879004	-0.5490981	-2.6890510
3	1.4720955	-0.1851853	-0.1252224	0.1019530	1.0006973	0.1254696	-2.6890510
4	1.4720955	0.4533848	-0.1252224	0.7391590	0.3879004	0.1254696	-0.2437182
5	1.4720955	-0.1851853	-1.4433524	1.3763651	-0.8376933	-1.2236657	-0.2437182
6	-1.7142584	-2.1008956	-0.7842874	-1.1724591	-1.4504902	0.1254696	-1.0588291
7	1.4720955	1.7305249	-0.7842874	1.3763651	-0.8376933	-1.8982334	-0.2437182
8	1.4720955	0.4533848	1.1929077	0.7391590	-1.4504902	-1.2236657	-1.0588291
9	0.8348247	0.4533848	1.1929077	-0.5352531	0.3879004	-0.5490981	0.5713928
10	0.1975539	1.0919548	1.1929077	-0.5352531	0.3879004	0.8000372	-1.0588291

DESCRIPTION

Standaardization of variables are set to zero mean and unit variance to ensure equal contribution to clustering

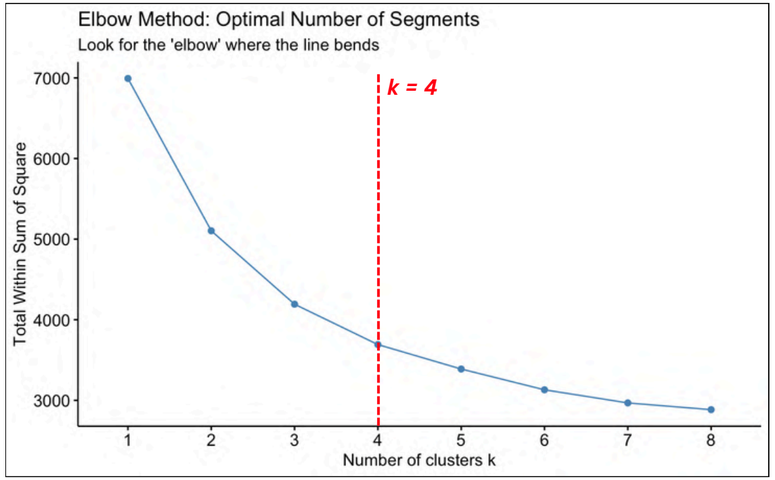
Marketing Analysis

Data Processing and Analysis

2 Elbow Plots + Dendrogram

Diagnosis 1

Elbow Method on Scaled Data

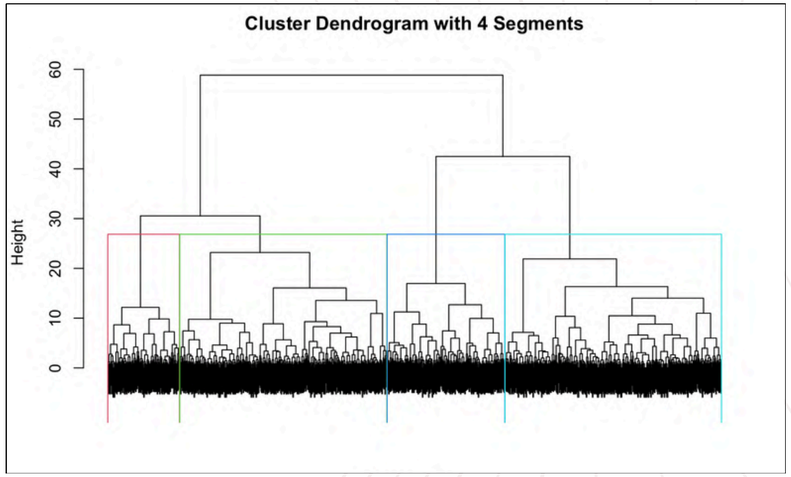


DESCRIPTION

- The bend occurs at k = 4.
- After k = 4, the drop in the WSS becomes marginal.
- Adding more clusters produces diminishing returns.
- Based on the analysis, the first diagnostic suggests that the optimal segments is k = 4.

Diagnosis 2

Dendrogram with 4 boxes

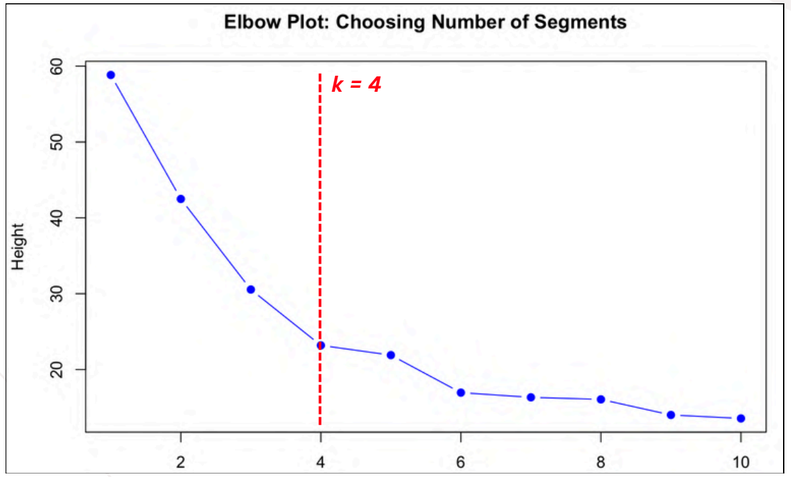


DESCRIPTION

- A dendrogram is applied as an independent methodological check as a second diagnostic.
- A different approach from the k-means, as this approach builds the clusters from the bottom up.
- Four cleanly separated branches confirm k = 4.
- Supporting merge-height plot points to k = 4.

Diagnosis 3

Hierarchical merge heights



DESCRIPTION

- Height drops significantly from k = 1 to k =4.
- After k = 4 merge heights flattens significantly.
- The dendrogram and merge heights result to the same hierarchical clustering.
- This approach is to provide numerical confirmation of the four branches seen in the dendrogram.

The diagnostics conducted concluded on the convergence on k = 4. This gives strong confidence in our segmentation solution

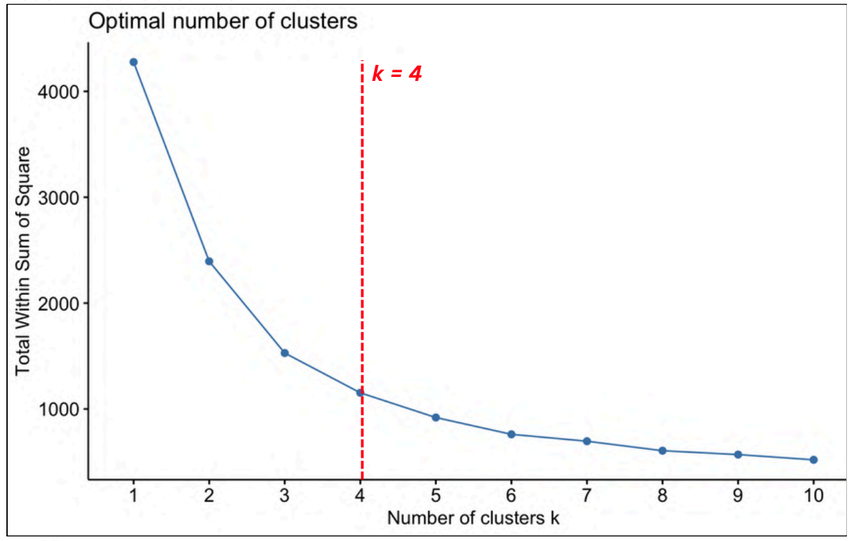
Marketing Analysis

Data Processing and Analysis

3 Principal Component Analysis (PCA)

It's a statistical technique that **reduces dimensions** in large datasets to principal components, retaining most original information and **transforming correlated variables into smaller sets.**

Elbow Method on PCA Data



DESCRIPTION

Why PCA before clustering?

1. ***Multicollinearity:***
   a. The dataset may include correlated variables that would be double-counting in the k-means.
2. ***Clearer Cluster Separation:***
   b. To minimise or prevent noises or distorted distances produced by the 7 dimensions.
3. ***Visualisation:***
   c. Combining the variables into new dimensions (PC1 + PC2) reveals a clearer, more distinctive cluster.
4. ***Shaper Elbow:***
   d. Utilizing the two new axes (the principal components), the bend at k = 4 is more defined on PCA data than raw data.

Variance Table

Component	Variance Explained	Cummulative
PC1	38.20%	38.20%
PC2	23.00%	61.20%

DESCRIPTION

What are the key insights gained?

1. PC1 + PC2 retained 61.2 % of the total variance in the original 7 variables, which is sufficient for robust segmentation.

2. Final k-means solution run on PCA data rather than the standardised variables produced a cleaner and better - separated clusters.

3. The elbow on PCA data produced a sharper bend at k = 4.

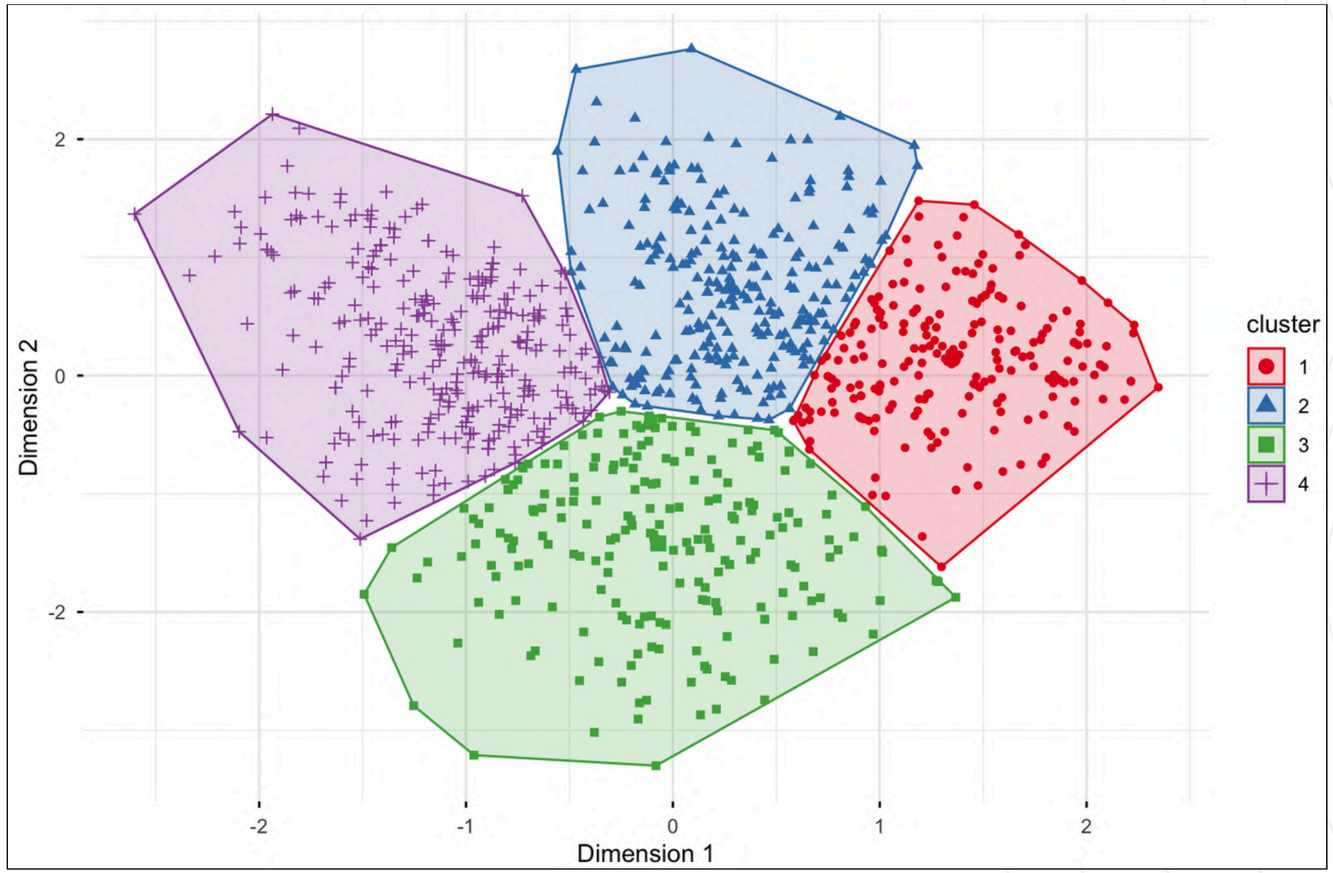


Marketing Analysis

Data Processing and Analysis

4 Cluster

Elbow Method on Scaled Data



1

Healthy Professionals

n = 216, 21.6%

- Young, urban, well-educated professionals.
- Highest income group (majority postgraduate).
- Predominantly female; technical savvy
- Smartwatch as a **lifestyle extension** (ex. wellness, task automation, and style)
- Willing to pay more for design, quality, and features

2

Fitness Enthusiasts

n = 258, 25.8%

- Active, sport-oriented lifestyle (runners, cyclists, gym)
- Uses smartwatches as **performance equipment**
- Athletic tracking as a priority, followed by wellness
- Not focused on style, notifications, or task automation
- Middle-income, willing to buy if the watch improves training

3

Busy Professionals

n = 214, 21.4%

- Older professionals (ex. late 30s, mid-career)
- **Demands real-time update** (emails, calendar, traffic, news)
- Health and athletics features are not a priority
- Gender balanced

⚠ **Already well served by Apple & Samsung**

4

Disengaged Users

n = 312, 31.2%

- Largest group but **least engaged segment**
- Lower income, older
- No real preference for any smartwatch feature
- Consistently have low interest in every attribute

Marketing Analysis

Data Processing and Analysis

5 Testing for Significance

Variable	F_value	p_value	eta_squared	Effect_size	Levene_p	Levene_warn	p_bonferroni	Sig_after_bonf
Wellness	592.8	5.493E-221	0.641	Large	1.033E-17	Unequal var - use Welch	4.944E-220	***
Style	482.76	1.321E-193	0.593	Large	0.001317	Unequal var - use Welch	1.189E-192	***
Athlete	422.48	5.421E-177	0.56	Large	1.832E-11	Unequal var - use Welch	4.879E-176	***
TaskMgm	212.74	1.266E-106	0.391	Large	2.425E-32	Unequal var - use Welch	1.139E-105	***
Age	210.95	6.531E-106	0.389	Large	7.492E-22	Unequal var - use Welch	5.878E-105	***
TimelyInf	202	2.534E-102	0.378	Large	8.939E-10	Unequal var - use Welch	2.281E-101	***
DeviceSt	177.68	2.917E-92	0.349	Large	0.01282	Unequal var - use Welch	2.625E-91	***
ConstCom	150.4	2.18E-80	0.312	Large	0.0002187	Unequal var - use Welch	1.962E-79	***
Income	52.32	2.106E-31	0.136	Medium	1.018E-07	Unequal var - use Welch	1.895E-30	***

	A	B	C	D	E	F	G
1	Variable	Pair	diff	lwr	upr	p adj	Sig
2	ConstCom	2-1	-1.33	-1.64	-1.02	0	***
3	ConstCom	3-1	-0.17	-0.49	0.16	0.5495	ns
4	ConstCom	4-1	-2.1	-2.39	-1.8	0	***
5	ConstCom	3-2	1.16	0.85	1.47	0	***
6	ConstCom	4-2	-0.77	-1.05	-0.49	2.411E-11	***
7	ConstCom	4-3	-1.93	-2.23	-1.63	0	***
8	TimelyInf	2-1	-1.13	-1.42	-0.84	0	***
9	TimelyInf	3-1	0.79	0.49	1.1	2.796E-10	***

Test 1

One-Way ANOVA

What it tells?

Whether the cluster means are significantly different across 4 segments for each of the 7 attributes.

The p-value alone is insufficient - η2 is used

- All 7 attributes show a large effect.
  - η2 >= 0.14
  - Wellness with *η2 = 0.641 - cluster, explains 64% of all wellness variation*

Test 2

Bonferroni Correction

What it tells?

Controls the false positive rate across 7 tests.

- All variables remain significant after correction.
- p < 0.001 (***) for all 7.
- Concludes that the results are not attributable to noise

Test 3

Levene's Test + Welch ANOVA

What it tells?

Checks the equal variance assumption of standard ANOVA - Welch ANOVA

- F-statistic is higher under Welch
- Wellness: 592.80 >>> 657.45
- Results are robust regardless of which test applied.

Test 4

Tukey HSD

What it tells?

Pairwise comparison across every cluster combination.

- Clean separation on almost every variable.
- Non-insignificant pairs are informative.
- Athlete tied s1- s2, Style tied s3 - s4.

Marketing Analysis

Competition Analysis: Evaluation Framework

Segment Attractiveness

The potential appeal of a specific segment to a business helps businesses to prioritize for profitability and success.
(Fiveable, 2025)

5 Criteria: How attractive is each segment?

1. **Segment Size**
   The number of customers or market share percentage
2. **Income Level**
   The ability to purchase a smartwatch: Mean Income Score
3. **Feature Engagement**
   The strength of smartwatch feature demand: Mean Attribute Score
4. **Aetna Reachability**
   Aetna's ability to distribute through Aetna's 70+ corpoerate wellness prgrammes
5. **Growth Potential**
   Market opportunity in the corporate wellness

Intel's Competitve Strength

Is Intel's capabilities and advanatges they can perform in each segment

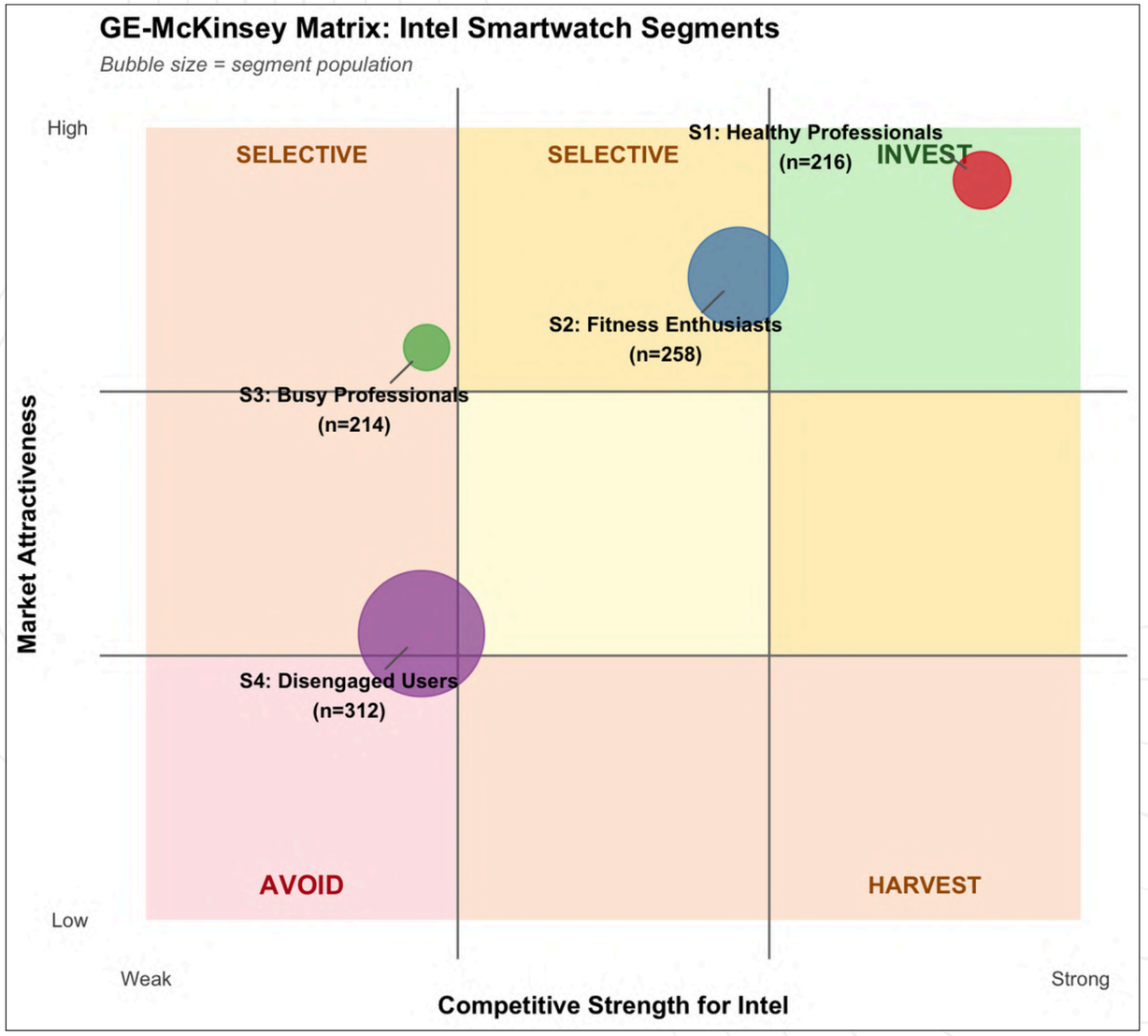
4 Dimensions: Quantifying Intel's ability to compete

1. **Sensor Technology**
   The number of customers or market share percentage
2. **Partner Ecosystem Advantage**
   Microsoft, Aetna, and CVS Health partnerships
3. **Competitive Landscape**
   Scale of competitor market share
4. **Market Gap Fulfillment**
   The ability to meet or address unmet needs

Marketing Analysis

Competition Analysis: GE-Matrix

GE - Matrix



Healthy Professionals

INVEST

- Highest Income (3.77) + Strongest Wellness Engagement (6.22)
- High reachability through Aetna's Corporate Programes

aetnaCVSHealth

Busy Professionals

SECONDARY

OPTION 2

- Strong Athletic (5.04) + Wellness (5.23)
- Limited Aetna reachability, due to performance-driven

GARMIN

Fitness Enthusiasts

AVOID

- TimelyInf (5.73) and ConstCom (5.56) = **'Communication-Centric'**
- ***Competitors already serve***

WATCH SAMSUNG Galaxy Watch

Disengaged Users

AVOID

- All scores below 3.7 + low income (2.73)
- No needs or desires for Intel to address

Marketing Analysis

Competition Analysis: Perceptual Map

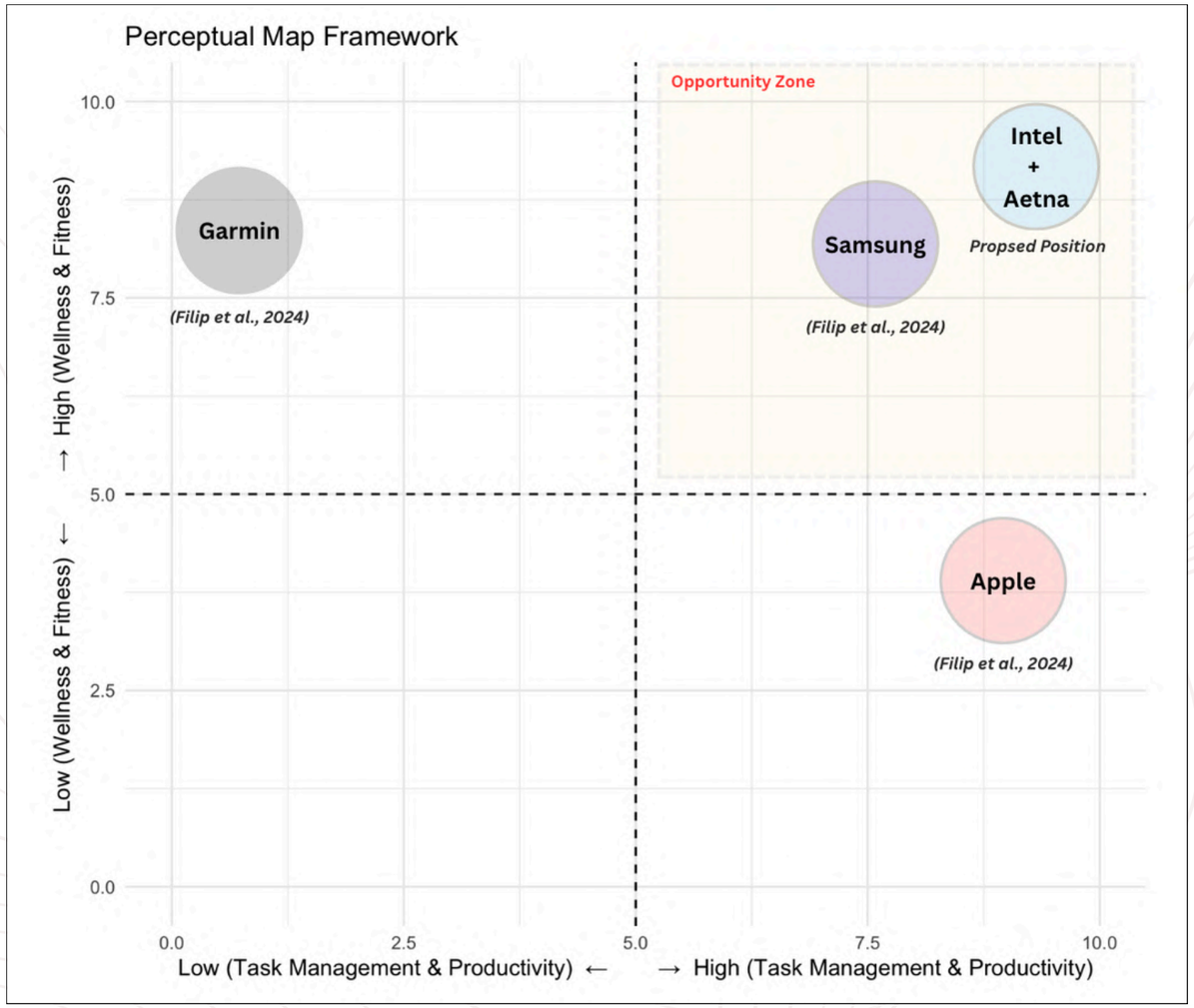
High W&F + Low T&P

GARMIN

- Athlete Specialist; Highest wellness orientation.
- Limited to none Productivity integration
- Low Threat to Intel + Aetna postioning

Abbreviations

1. **W&F** Wellness + Fitness
2. **T&P** Task Management + Productivity



Note: Competitor positions informed by independent brand perception research (Filip et al., 2024). Intel/Aetna position represents a strategic aspiration based on target segment priorities and proposed partnership capabilities.

High W&F + High T&P

intel + aetna

- Opportunity Zone: Proposed Position
- Insurance-backed wellness through Aetna
- Microsoft 365 productivity integration

SAMSUNG Galaxy Watch

- Main Competitor (balanced positioning)
- Health Partnerships
- B2C oriented

Low W&F + High T&P

WATCH

- Smartphone Ecosystem Dominant: High productivity.
- Moderate Wellness
- Majority serves professionals not Healthy Professionals

Marketing Analysis

Partner Recommendation

Amazon

“to be Earth's most customer-centric company”

- Discontinuation of the Halo Division and abandonment of the health wearable hardware entirely (Reuters, 2023).
- Limited to strategic Interest - No longer active in the health wearable sector.
- Alexa Integration is insufficient for the wellness segment.

Google

"to provide access to the world's information in one click."

Direct Competition



- Commercial Presence in the same industry as the proposed product.
- The partnership would create a direct conflict of interest.

Aetna

"To help people achieve health and financial security by providing easy access to safe, cost-effective, high-quality health care and protecting their finances against health-related risks."

+70 Corporate Wellness Programs - direct B2B Employer Access

9000 + CVS retail locations - Secondary distribution to B2C

Health 100 - AI Health tracking and biometric analytics (CVS, 2026)

Marketing Analysis

Go-to-Market Strategy

B2B - Employer Wellness via Aetna

Primary Channel



- Intel watches are bundled as a health incentive within corporate benefits packages
- Aetna's 70+ existing corporate wellness programmes provide immediate B2B reach
- Device co-financed or subsidised as a corporate wellness asset, reduces the price barrier
- Targets Healthy Professionals directly through the employer, no retail competition needed

B2C - CVS Health Retails

Secondary Channel



- 9,000+ CVS pharmacy locations across the United States
- Targets consumers outside employer programmes direct retail access
- CVS ExtraCare loyalty programme supports repeat engagement
- Established e-commerce and nationwide medical supply chain infrastructure

Insurance Credit

Aetna legitimises health data handling and wellness claims — differentiated from consumer tech competitors

VS

Avoids Apple Competition

B2B first strategy bypasses direct retail competition with Apple's premium consumer dominance

Sustainable Revenue

Employer-subsidised model builds recurring B2B revenue while CVS retail captures individual consumers

Implementation Roadmap

Phased market entry strategy for Intel + Aetna/CVS partnership

1

B2B LAUNCH

Months 1–6

- → Aetna's 70+ corporate programs
- → Build healthcare credibility
- → Collect real health data

2

B2C EXPANSION

Months 6–12

- → CVS Health retail (9k+ stores)
- → Consumers outside employer programs
- → Leverage CVS supply chain

3

SCALE & OPTIMIZE

Month 12+

- → Optimize based on data
- → Expand distribution channels
- → Sustain recurring revenue

De-risked GTM Strategy

PARTNERSHIP ECOSYSTEM

AETNA → INTEL → CVS

Credibility → Technology → Distribution

ADDRESSABLE MARKET

47.4%

PRIMARY SEGMENTS

Healthy Professionals + Fitness Enthusiasts

DISTRIBUTION

Aetna + CVS Health

Marketing Positioning & Messaging

Targeted value propositions by segment



Healthy Professionals

21.6% | High income, health-conscious

KEY MESSAGE

"Stay healthy. Stay productive. Backed by insurance science."

VALUE PROPOSITION

- Real-time health insights
- Employer subsidy (premium access)
- Wellness rewards integration

CHANNELS

Employer HR, LinkedIn, wellness platforms

Fitness Enthusiasts

25.8% | Performance-driven athletes

KEY MESSAGE

"Train smarter. Perform better. Data-driven edge."

VALUE PROPOSITION

- Advanced athletic metrics (HR, recovery)
- Military-grade durability
- Employer wellness accessibility

CHANNELS

Strava, fitness sponsorships, athletic communities



Combined reach: 47.4% addressable market via single go-to-market (Aetna B2B + CVS retail). Intel differentiates as healthcare-credible wellness partner—the only company offering health-first positioning + employer accessibility + premium quality.